

Lab: Actions Speak Louder -- The Explore vs. Exploit Challenge

Objective

Learning Goal: Investigate how you choose between exploring new options and sticking with what works, and discover how actions reduce uncertainty.

This is a **group challenge** designed for middle school students (Grades 6-8).

Materials Needed

- 4 paper bags or cups (labeled A, B, C, D)
 - Colored tokens or candies (different values in each bag)
 - Score sheet and pencil
 - Lab journal or notebook
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Part 1: The Mystery Bag Game (15 min)

Your teacher has set up 4 bags. Each bag contains tokens worth different point values, but you do not know which bag is best. You get 10 turns total. Each turn, you pick ONE bag and draw a token.

The dilemma: Do you keep picking from a bag that gave good results (exploit), or try a new bag to see if it is better (explore)?

Turn	Bag Chosen (A/B/C/D)	Points Earned	Why did you choose this bag?	Explore or Exploit?
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
Total				

Write your response here

Part 2: Action as Uncertainty Reduction (10 min)

In Active Inference, actions do not just get you rewards -- they also **reduce uncertainty**. Sometimes the best action is one that gives you information, even if it does not pay off right away.

Think about these real-life scenarios. For each one, decide: is the best action to explore (get information) or exploit (use what you already know)?

Scenario	Explore or Exploit?	Why?
You have a big test tomorrow and know one study method that works OK		
You are choosing a new elective and have never tried art or coding		
Your usual route to school is blocked by construction		
You found a great YouTube channel for homework help		
You are at a new restaurant with a huge menu		

Write your response here

Part 3: Action Sequence Challenge (10 min)

Most real actions are not one-time choices -- they are sequences (policies). Map out an action sequence for one of these goals:

Choose one: - Getting a better grade on your next math test - Making a new friend this month
- Learning a new skill (instrument, sport, cooking)

Step #	Action	What information does this give you?	What uncertainty does it reduce?
1			
2			
3			
4			
5			

Write your response here

Part 4: Reflection Table

Question	Your Answer
What is the difference between exploring and exploiting?	
Why do actions reduce uncertainty?	
What is a policy (action sequence)?	
When in your life do you explore too little?	
How does Active Inference explain why we take actions?	

Write your response here